Group	Method	Parameter Options	Source besides Scikit-learn
Missing Value Imputation	Mean	None	
Missing Value Imputation	Median	None	
Missing Value Imputation	Most Frequent	None	
Missing Value Imputation	knn-Imputer	{n_neighbors: [1, 16)}	Troyanskaya et al. [9]
Outlier Removal	LocalOutlierFactor	{p_neighbors*: {0.01, 0.05, 0.1, 0.2, 0.3, 0.5}, algorithm: {auto, ball_tree, kd_tree, brute}, p_leaf_size: {1, 5, 10, 20, 30, 50}, p: {1, 2, 5, 10}, contamination: {auto, 0.05, 0.1, 0.2, 0.3, 0.4}	Breunig et al. [1]
Outlier Removal	IsolationForest	{n_estimators: {10,50,100,200,500}, max_features: {2, 5, 10, 25, 35}, bootstrap: {True, False}, warm_start: {True, False}	Liu et al. [4]
Outlier Removal	OneClassSVM	{kernel: {linear, poly, rbf, sigmoid}, gamma: {scale, auto}, nu: {0.3, 0.5, 0.7}, shrinking: {True, False}, max_iter: {100, 500, 1000}	
Outlier Removal	EllipticEnvelope	{assume_centered: {True, False}, contamination: {0.1, 0.2, 0.3, 0.4, 0.5}	Rousseeuw and Driessen [6]
Outlier Removal	SGDOneClassSVM	{learning_rate: {constant, optimal, invscaling, adaptive}, warm_start: {True, False}, nu: {0.3, 0.5, 0.7}, max_iter: {100, 500, 1000}, eta0: {0.1, 0.2, 0.3}	
Scaling	PowerTransformer	{standardize: {True, False}	Yeo and Johnson [10]
Scaling	RobustScaler	{with_centering: {True, False}, with_scaling: {True, False}, with_variance: {True, False}	
Scaling	StandardScaler	{with_mean: {True, False}, with_std: {True, False}	
Scaling	MinMaxScaler	{None}	
Scaling	QuantileTransformer	{p_quantiles: {0.3, 0.5, 0.7, 1}, output_distribution: {uniform, normal}	
Scaling	MaxAbsScaler	{None}	
Dimensionality Reduction	pca	{n_components: special treatment, whiten: {True, False}, svd_solver: {auto, full, arpack, randomized}, power_iteration_normalizer: {auto,QR, LU}, max_iter: {100, 500, 1000}	Minka [5], Tipping and Bishop [8], and Halko et al. [3]
Dimensionality Reduction	GaussianRandomProjection	{n_components: special treatment, eps: {0.01, 0.1, 0.3}	
Dimensionality Reduction	LocallyLinearEmbedding	{n_components: special treatment, p_neighbors*: {0.01, 0.05, 0.1}, eigensolver: {auto, arpack, dense}, max_iter: {50, 100, 500, 1000}, method: {standard, hessian, modified, ltsa}, neighbors_algorithm: {auto, brute, kd_tree, ball_tree}	Roweis and Saul [7], Donoho and Grimes [2], Zhang and Wang [11], and Zhang and Zha [12]

Table 2: Preprocessing Methods and their Hyperparameters

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